

LONGEST SUBSTRING WITHOUT REPEATING CHARACTERS

Java Programming Assignment

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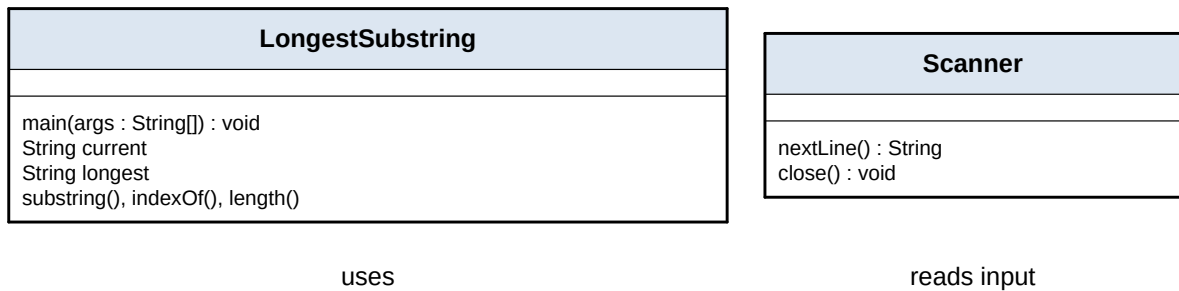
1. ALGORITHM

Steps for executing the LongestSubstring program:

Step	Algorithm
1	Start the program.
2	Import the Scanner class.
3	Create a Scanner object to read a string from the user.
4	Initialize two strings: current = "" and longest = "".
5	Traverse the input string character by character using a for loop.
6	Extract the current character using substring(i, i + 1).
7	Check whether the character already exists in current using indexOf().
8	If it exists, remove the characters before and including its previous occurrence using substring(pos + 1).
9	Append the current character to current.
10	If current is longer than longest, update longest with current.
11	After the loop, display the longest substring and its length.
12	Close the Scanner and stop the program.

2. UML CLASS DIAGRAM

The LongestSubstring class uses the main() method to process the input string with String operations.



UML: LongestSubstring is the main class. It uses Scanner for input and String operations to find the longest substring without repeating characters.

3. JAVA SOURCE CODE

File name: **LongestSubstring.java**

Note: The final line is corrected to `sc.close();` so the program compiles.

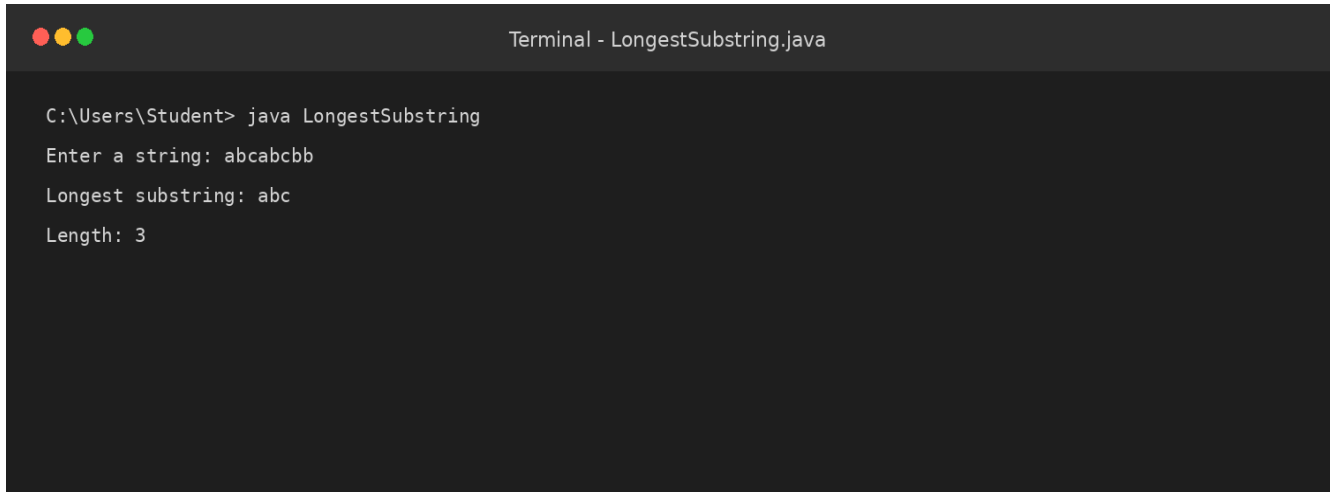
```
import java.util.Scanner;
public class LongestSubString {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = sc.nextLine();
        String current = "";
        String longest = "";
        for (int i = 0; i < str.length(); i++) {
            String ch = str.substring(i, i + 1);
            int pos = current.indexOf(ch);
            if (pos != -1) {
                current = current.substring(pos + 1);
            }
            current = current + ch;
            if (current.length() > longest.length()) {
                longest = current;
            }
        }
    }
}
```

3. JAVA SOURCE CODE (CONTINUED)

```
System.out.println("Longest substring: " + longest);  
System.out.println("Length: " + longest.length());  
sc.close();  
}  
}
```

4. OUTPUT SCREEN - VS CODE STYLE

Example execution for the input **abcabcbb**:

A screenshot of a terminal window with a dark background. The title bar at the top reads "Terminal - LongestSubstring.java" and features three colored window control buttons (red, yellow, green) on the left. The terminal content shows a command prompt session: the user enters "C:\Users\Student> java LongestSubstring", followed by the program's prompts "Enter a string: abcabcbb", "Longest substring: abc", and "Length: 3".

```
Terminal - LongestSubstring.java

C:\Users\Student> java LongestSubstring
Enter a string: abcabcbb
Longest substring: abc
Length: 3
```

Result: Longest substring = **abc**, Length = **3**.